Lecture 01 — Why linear algebra? Vectors as representations

Mon Aug 24

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Last updated: August 18, 2026.

Reading: Boyd 1.1. Related objectives: Lectures. Schedule.

Bring paper. We will fill the blanks together, work one or two examples on the board, and leave time for **Your turn**.

Learning objectives

By the end of class, students should be able to:

1. Identify objects that can be represented as vectors.
2. Interpret vector entries, units, and indexing in context.
3. Distinguish scalars from vectors.
4. Translate a small data, position, signal, or measurement problem into vector form.

Fill in

An *n-vector* is an ordered list of ______________ numbers, written

$$x = (x_1, x_2, \dots, x_n).$$

The number n is the ______________ (or dimension) of x. Each x_i is an ______________ of x. In this course we index from $i = 1$ to $i = n$ (Boyd's convention).

Two n-vectors a and b are equal when ___________________.

A *scalar* is ___________________. The number x_3 is a ____________; the list x is a ____________.

The *zero vector* 0 has ______________. The *ones vector* $\mathbf{1}$ has ______________. The *standard unit vector* e_i has _____________________.

In an application, we often keep a separate list of ______________ so that we know what each entry means and what ____________ it has.

Worked in class

W1. An Arcata weather station records four successive high temperatures, in °F:

$$t = (63, 59, 61, 66).$$

What is n? What is t_3, and what does it mean in context? Is 63 a vector or a scalar? Could you add a fifth day without changing the meaning of $t_1, \dots, t_4$?

W2. A redwood seedling is described by height in cm, basal diameter in mm, and soil moisture in percent. Write a 3-vector f for a seedling that is 18 cm tall, 4 mm across, and 27% moisture. Give an entry label for each index. Which entries are allowed to be compared as "the same kind of quantity," and which are not?

Your turn

Y1. An RGB color is the 3-vector $c = (0.15, 0.80, 0.35)$, with entries between 0 and 1 for red, green, and blue. Which entry is largest, and what color does that suggest? What 3-vector is black in this encoding?

Y2. A short field note is scored against the dictionary (fog, rain, wind). The word-count vector is $w = (7, 2, 0)$. What is n? What does w_1 mean? What does $w_3 = 0$ tell you?

Answers (Your turn only)

Y1. The second entry is largest, so the color is mostly green. Black is $(0, 0, 0)$.
Y2. $n = 3$. The word *fog* appears 7 times. *Wind* does not appear.